

Skill Check 1: Basics of R

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human. Cell phones are to be handed in to the instructor and all other digital devices must be stored in bags underneath the bench or in the lab cubbies.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: How Computers Work

People code in languages using letters, numbers, and symbols. What type of code translates our code into something computers can understand? Why is this necessary?

Question 2: Debugging and Troubleshooting

The following line of code does not work:

```
df <- data.frame("words" = c("apple", "creature", "puppy") "actions" = 1:3, )
```

It produces the following error:

```
Error: unexpected string constant in "df <- data.frame("words" = c("apple",  
"creature", "puppy") "actions"
```

Rewrite the line of code that corrects the error and runs successfully below.

Question 3: Data Frames

Write a line of code that calculates the mean uptake of CO₂ for plant QN2 at concentration 175 in the C02 data set.

☑ C02	84 obs. of 5 variables	
\$ Plant	: Ord.factor w/ 12 levels "Qn1"<"Qn2"<"Qn3"<...	
\$ Type	: Factor w/ 2 levels "Quebec","Mississippi": ...	
\$ Treatment	: Factor w/ 2 levels "nonchilled","chilled": ...	
\$ conc	: num 95 175 250 350 500 675 1000 95 175 250...	
\$ uptake	: num 16 30.4 34.8 37.2 35.3 39.2 39.7 13.6 ...	
- attr(*, "formula")	=Class 'formula' language uptake ~ ...	
.. ..- attr(*, ".Environment")	=<environment: R_EmptyEnv...	
- attr(*, "outer")	=Class 'formula' language ~Treatment ...	
.. ..- attr(*, ".Environment")	=<environment: R_EmptyEnv...	

Question 4: Matrices and Arrays

Write a line of code that creates a matrix which has 80 columns and 60 rows using `sample()` to generate random integers for its data.

Question 5: Lists

I have created a list stored with the name `my_list` with the following code:

```
my_list <- list("blep" = c("a", "b", "c", "d", "e"),
               "matt" = matrix(runif(500), ncol = 50, nrow = 100),
               "lindsay" = data.frame("x" = c(1.1, 1.2, 1.3),
                                       "y" = c(2.3, 5.6, 6.2)),
               10:100)
```

- a. How many members does this list contain?
- b. What data class is each member?
- c. How would you reference the fifth row and second column in the matrix in `my_list`?

Question 6: Class Coercion

If you run the following lines:

```
a <- c(0, 1, 0, 0)
any(a)
```

You will get a warning, but `any()` will return `TRUE`. Explain why this works, even though `any()` can only take logical values.

Question 7: Data Classes and Special Values

Is the following is a valid vector in R? Explain why or why not.

```
vec <- c(Inf, Inf, -Inf)
```

Question 8: Factors

Write a line of code that creates a factor with at least three levels that describes the color of at least four pairs of shoes you own or have owned.

Question 9: Vectors

What is an example of a function that can be used to produce a vector? Describe what this function does and/or produces. Give a working example.